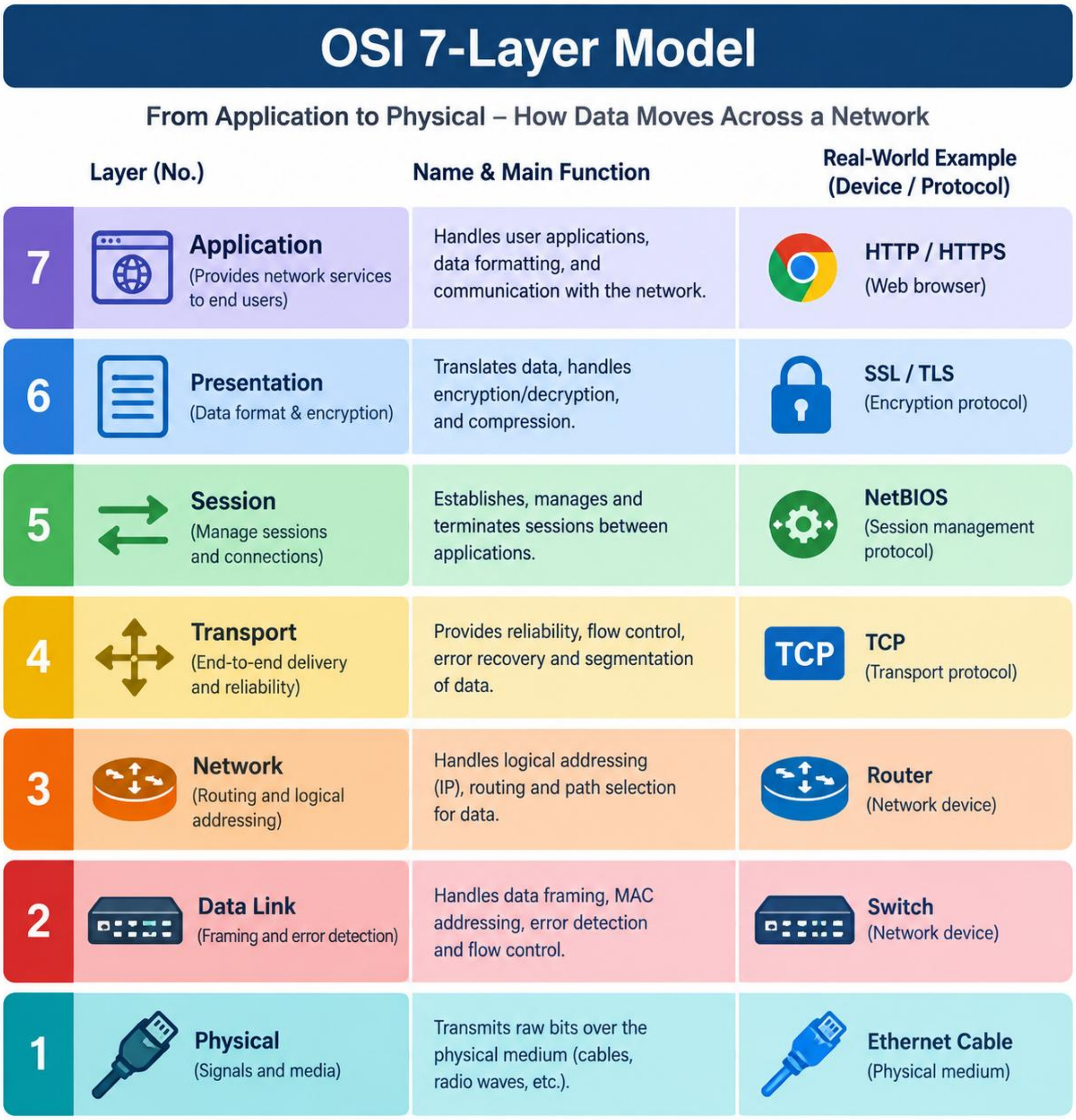
OSI Layer

Task (1):



OSI 7-Layer Model

From Application to Physical – How Data Moves Across a Network

Layer (No.)		Name & Main Function	Real-World Example (Device / Protocol)
7	Application (Provides network services to end users)	Handles user applications, data formatting, and communication with the network.	HTTP / HTTPS (Web browser)
6	Presentation (Data format & encryption)	Translates data, handles encryption/decryption, and compression.	SSL / TLS (Encryption protocol)
5	Session (Manage sessions and connections)	Establishes, manages and terminates sessions between applications.	NetBIOS (Session management protocol)
4	Transport (End-to-end delivery and reliability)	Provides reliability, flow control, error recovery and segmentation of data.	TCP (Transport protocol)
3	Network (Routing and logical addressing)	Handles logical addressing (IP), routing and path selection for data.	Router (Network device)
2	Data Link (Framing and error detection)	Handles data framing, MAC addressing, error detection and flow control.	Switch (Network device)
1	Physical (Signals and media)	Transmits raw bits over the physical medium (cables, radio waves, etc.).	Ethernet Cable (Physical medium)

Task (2):

Scenario	Correct OSI Layer
a) Encrypting data before sending	Layer 6 – Presentation
b) Routing packets across networks	Layer 3 – Network
c) Converting data to electrical signals	Layer 1 – Physical
d) Opening a web page in your browser	Layer 7 – Application
e) Detecting and correcting errors in frames	Layer 2 – Data Link

Task (3):

Example: WhatsApp

When I send a message on WhatsApp, several OSI layers work together:

1. **Layer 7 – Application:** WhatsApp handles the message and provides the user interface for sending/receiving it.
2. **Layer 4 – Transport:** TCP/UDP helps transport the data between my phone and the WhatsApp server.
3. **Layer 3 – Network:** IP is used to route the data packets from my phone to the destination/server.
4. **Layer 2 – Data Link:** Wi-Fi or mobile network transfers the frames between nearby network devices.
5. **Layer 1 – Physical:** Radio signals carry the data between my phone and the Wi-Fi router or mobile tower.

Task (4):

If you can ping the website's IP address successfully but cannot open the website, the problem is likely at Layer 7 (Application) or Layer 4 (Transport).

- Layer 7 – Application: The web service (HTTP/HTTPS) may be down, blocked, or misconfigured.
- Layer 4 – Transport: TCP port 80 (HTTP) or 443 (HTTPS) may be blocked by a firewall or the server may not be accepting connections.

Reason: Successful ping shows that the Network Layer (Layer 3) and basic connectivity are working. Therefore, the issue is likely with the web service or the ports used to access it.